

Sharpened Visual Memory Representations Are Reflected in Inferotemporal Cortex

 Barnes G. L. Jannuzzi,  Catrina M. Hacker,  Simon Bohn,  Travis Meyer, Madison L. Hay, and  Nicole C. Rust

Department of Psychology, University of Pennsylvania, Philadelphia, Pennsylvania

Humans and other primates can robustly report whether they have seen specific images before, even when those images are extremely similar to ones they have previously seen. Multiple lines of evidence suggest that pattern separation computations in the hippocampus (HC) contribute to this behavior by shaping the fidelity of visual memory. However, unclear is whether HC uniquely determines memory fidelity or whether computations in other brain areas also contribute. To investigate, we recorded neural signals from inferotemporal cortex (ITC) and HC of two rhesus monkeys (1 male, 1 female) as they performed a memory task in which they judged whether images were novel or exactly repeated in the presence of visually similar lure images with a range of visual similarities. We found behavioral evidence for sharpening, reflected as memory performance that was nonlinearly transformed relative to a benchmark defined by visual representations in ITC. As expected, we found that behavioral sharpening aligned with visual memory representations in HC. Surprisingly, and unaccounted for by HC pattern separation proposals, we also found neural correlates of behavioral sharpening reflected in ITC. These results, coupled with further analysis of the data, suggest that ITC contributes to shaping the fidelity of visual memory in the transformation from visual processing to memory storage and signaling.

Key words: familiarity; hippocampus; inferior temporal cortex; inferotemporal cortex; medial temporal lobe; recognition memory

Significance Statement

Visual recognition memories are stored with remarkable visual fidelity, allowing humans and other primates to distinguish images they have encountered from visually similar images they have not. This fidelity has long been attributed to computations in the hippocampus that sharpen visual representations before memory storage (“pattern separation”). Unclear is how this proposal aligns with other evidence that visual memories are stored within high-level visual cortex itself, before signals reach the hippocampus. Here we demonstrate that, like the hippocampus, inferotemporal cortex also reflects sharpened visual memory representations, suggesting that visual cortex contributes to shaping the visual fidelity of visual memory.

Introduction

Humans (and other primates) have a remarkable ability to remember the images they have seen. Subjects can correctly report whether they have seen specific images after seeing thousands of them, each only once and for a few seconds (Standing, 1973; Brady et al., 2008). In addition, subjects typically remember images with impressive visual detail, even in the absence of

specific instructions to do so (Standing, 1973; Vogt and Magnussen, 2007; Brady et al., 2008; Konkle et al., 2010; Stark et al., 2015, 2019). How do our brains manage to support the astounding visual specificity of visual recognition memory?

Several lines of evidence suggest a prominent role of the hippocampus (HC) in shaping the fidelity of visual memory. When visual memory is tested in patients with HC damage, they have preserved memory for visually distinct images, but they falsely remember lure images that are similar to those already seen at higher rates than controls (Stark et al., 2013, 2015, 2019). Similarly, individuals with an array of HC-implicated disorders display impaired memory performance for similar images, including patients with age-related mild cognitive impairment (Stark et al., 2013; Bennett et al., 2019) and Alzheimer’s disease (Ally et al., 2013; Monti et al., 2014). Likewise, fMRI comparisons of visual representations between the HC and a primary input, entorhinal cortex, suggest that hippocampal memory representations are more sparse, implying computations within the hippocampus sharpen visual representations (Bakker et al., 2008; Lacy et al., 2011; Berron et al., 2016).

Received April 28, 2025; revised Sept. 26, 2025; accepted Oct. 26, 2025.

Author contributions: B.G.L.J. and N.C.R. designed research; B.G.L.J., C.M.H., S.B., T.M., and M.L.H. performed research; B.G.L.J., C.M.H., S.B., and N.C.R. analyzed data; B.G.L.J. and N.C.R. wrote the paper.

This work was supported by the National Science Foundation (award 2043255 to N.C.R.), the National Eye Institute of the NIH (award R01EY020851 to N.C.R.), and the Simons Foundation (Simons Collaboration on the Global Brain award 543033 to N.C.R.).

The authors declare no competing financial interests.

Correspondence should be addressed to Nicole C. Rust at nrust@psych.upenn.edu.

This paper contains supplemental material available at: <https://doi.org/10.1523/JNEUROSCI.0833-25.2025>

<https://doi.org/10.1523/JNEUROSCI.0833-25.2025>

Copyright © 2025 Jannuzzi et al.

This is an open-access article distributed under the terms of the Creative Commons Attribution 4.0

International license, which permits unrestricted use, distribution and reproduction in any medium provided that the original work is properly attributed.

These results have led to the “pattern separation hypothesis” for how HC may shape visual memory by sparsifying visual representations. The notion is that visual representations of similar images in high-level visual cortex (including inferotemporal cortex (ITC)) are highly overlapping (Fig. 1, left), and because memory storage and signaling is a noisy, information-degrading process, these representations cannot support the visual specificity of visual memory (Yassa and Stark, 2011; Leal and Yassa, 2018; Stark et al., 2019). To resolve this downstream of ITC, pattern separation computations in HC transform overlapping and distributed representations arriving from entorhinal cortex into more sparsely distributed representations (Fig. 1, right). These dissimilar visual representations are then protected from confusion during noisy memory storage and signaling. Consistent with this proposal, one study found that visual memory behavior in monkeys could be decoded from HC (Sakon and Suzuki, 2019).

While evidence implicating HC is compelling, it seems to contradict evidence that visual memories are stored within high-level visual cortex, before signals reach HC. Namely, a number of studies have demonstrated an alignment between visual memory behavior and repetition suppression (RS), an adaptation-like response reduction that occurs in ITC upon repeating a stimulus (Fahy et al., 1993; Li et al., 1993; Miller and Desimone, 1994; Xiang and Brown, 1998; Meyer and Rust, 2018; Mehrpour et al., 2021), thereby supporting ITC RS as a candidate signal that the brain uses to drive visual memory. It is possible that fidelity-shaping computations within ITC also contribute to the transformation from vision to visual memory. However, ITC visual memory representations of highly similar images have never been systematically compared with the fidelity of visual memory behavior.

Here, we ask whether and how transformations between ITC visual representations and HC memory representations occur. We trained monkeys to perform a visual memory task in which they reported familiarity for novel and exactly repeated images, as well as visually similar lures. As monkeys performed this task, we recorded neural data from HC and ITC. We found robust behavioral evidence of sharpening, reflected as nonlinearly transformed memory performance on lure images relative to a model of ITC visual representations (Kubilius et al., 2019). As expected from previous work, we found that visual memory representations in HC aligned with visual memory behavior.

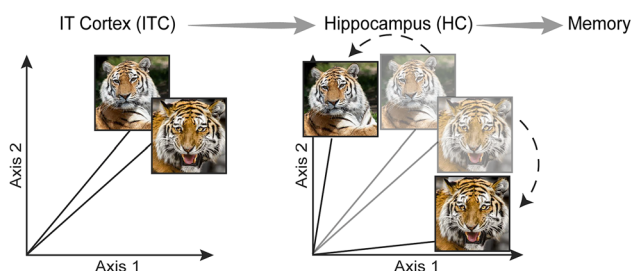


Figure 1. The hippocampal pattern separation hypothesis. Left, Hypothetical ITC population responses to two tiger images that are highly overlapping. Because of their high similarity, they have highly overlapping neural representations. The visual representations of these two images exist in high- d population space (dimensionality equal to the number of ITC neurons), but only two dimensions are visualized here. Because memory storage and signaling is a noisy process, confusion can follow from highly similar images like these, where an individual falsely reports seeing one image when in truth, they saw the other. Right, Hypothetical population responses to the same two images in HC following pattern separation. Because the images are less overlapping, memory is less susceptible to noise. Dotted arrows emphasize the transformation in the population representations between ITC and HC.

Surprisingly, we also found evidence of sharpening in ITC. These results, coupled with additional analyses, suggest that ITC contributes to shaping the fidelity of visual memory in the transformation from visual to memory representations.

Materials and Methods

Subjects

Experiments were performed on two adult rhesus macaque monkeys (one female, *Macaca mulatta*) with implanted headposts and recording chambers. All procedures were performed in accordance with the guidelines of the University of Pennsylvania Institutional Animal Care and Use Committee.

The modified mnemonic similarity task

All behavioral training and testing was performed using standard operant conditioning with juice as a reward. The monkeys' eyes were tracked using high-accuracy infrared video eye tracking under head-stabilized conditions using EyeLink software and hardware (<https://www.sr-research.com/>). Image stimuli were presented on an 85 Hz refresh rate LCD monitor using customized software (<http://mworks-project.org>).

Trials were initiated by the monkey fixating on a red fixation square (0.25°) at the center of a gray screen, within a square window of length 3.5°, followed by a 500 ms delay before a (4°) full-color image appeared. The monkey maintained fixation on the image for 500 ms before the red fixation square turned to a green go cue, and two targets were displayed 8° above and below the image. The monkey could then make a saccade to one of the two targets to indicate if that stimulus was novel (never before seen) or repeated (the exact same image, seen exactly once before). Images were never presented more than twice during the entire training and testing period of the experiment and fell into one of two categories. The first category was “exact repeats,” where an image was shown once as novel and once as repeated. The second category were “lure pairs” where, rather than being followed by a repeated presentation of an image, the image was followed by a “lure” image showing the a novel image of the same object at titrated levels of similarity to the novel image. Because the lure images had never been seen before, they were rewarded as novel images. To keep the ratio of novel versus repeated images approximately equal, the lure images were repeated (these trials are not included in the analyses we report here).

Saccades made before the go cues resulted in an aborted trial (and were excluded from the analyses we report here). The association between the target (up vs down) and the report (novel vs repeated) was swapped between the two animals. For Monkey 1, the stimulus remained on the screen until a saccade away from the stimulus was detected. For Monkey 2, the stimulus disappeared at the end of the 500 ms presentation.

In the experiment, the image repeats and lures were shown with an “ N -back” (N = number of interleaving images) delay adjusted to approximately equate performance between the two animals, determined to be ~ 24 images apart in Monkey 1 and ~ 8 images apart in Monkey 2. The target N -back was chosen such that performance for exact repeats was near 100% and performance on lure image pairs was off ceiling (and modulated by dissimilarity). N -backs for individual trials were randomly selected from a uniform distribution of nearby N -backs (Monkey 1: [23, 24, 25, 26], Monkey 2: [7, 8, 9, 10]). In addition, N -backs of [1, 2, 4] were shown throughout the experiment to serve both as fillers and as vigilance trials, and a small fraction of trials were filled with other N -backs as fillers (e.g., 3-back). These trials were not included in the analyses reported here.

In total, a behavior session included 1,500 trials. Of these, 480 were associated with experimental lures (160 novel images, 160 visually similar lures and 160 lure repeats), 900 were exact repeat images shown at matched n -backs to the lures (450 novel images and 450 repeats), and 120 were vigilance lures (40 novel images, 40 visually similar lures and 40 lure repeats).

Images

The images used in these experiments were gathered from five databases: the Ecocost image database (Mehrer et al., 2021), the Things image database (Hebart et al., 2019), the MemCat image database (Goetschalckx

and Wagemans, 2019), the LaMem image database (Khosla et al., 2015), and the MST task (Kirwan et al., 2007; Bakker et al., 2008). The vast majority of the images used in these experiments were Ecosem images. Images were cropped and resized to a square of dimension 256×256 pixels. Image memorability (Isola et al., 2011; Khosla et al., 2015) has been shown to affect the response magnitude to observed images (Jaegle et al., 2019). This poses a potential confound to interpreting magnitude-based memory signals. To control for this, images were run through an artificial network MemNet (Khosla et al., 2015) to yield a predicted memorability score for each stimulus. Because the memorability of an image is known to affect the magnitude of its neural response, and this project was not focused on disambiguating effects of image memorability, only high-memorability images with a MemNet score of 0.7–1 were considered for this project, and memorability distributions were matched between lure bins. Visual dissimilarity of lures was confirmed to not be predictable from low-level image statistics. For this analysis, a vector of 18 values for each image was computed to capture low-level image statistics, including six values (mean, variance, skew, kurtosis, max, and min) of the pixel values in each of the three color channels. Next, a leave-one-out, cross-validated linear correlation classifier was tested for its ability to classify which of three visual dissimilarity bins each test image belonged to based on these statistics. Performance of the classifier was at chance (32.7%, as compared with the chance benchmark of 33%).

Defining visual dissimilarity

There were two hurdles to overcome when defining visual dissimilarity. First, our experiment is designed to measure behavioral performance and its neural correlates relative to an ITC benchmark that we do not have direct access to before the experiment begins. Second, motivated by the robustness of single-trial familiarity behavior, our experiment focuses on the nature of familiarity representations in the brain following a single exposure, and the single-trial neural data we collect are insufficient to yield robust predictions of image dissimilarity for individual image pairs. To address these challenges, we turned to representational dissimilarity as defined by the dissimilarity of image representations in the brain-like artificial neural network CORnet_S-IT. CORnet_S-IT is a top-performing model evaluated for its ability to mimic the responses of ITC to arbitrary natural images in passively fixating monkeys, along with behavioral patterns of confusions in humans on an invariant object recognition task (Kubilius et al., 2019). Using methods described by Kriegeskorte et al. (2008), we created a representational $n \times n$ dissimilarity matrices, where n is the number of images within an object category, and each value is equal to 1 minus the Pearson's correlation of two images' representations in CORnet_S-IT. Of note, CORnet_S-IT model predictions are noiseless and thus the visual dissimilarity of an image to itself is 0. We expect dissimilarities of two nonpaired novel images to be ~ 1 . We verified this by computing the dissimilarity of each novel images to another (nonlure) novel image not confined to the same object category. This indeed resulted in a distribution approximated by 1.

From each image set, for each object category, a dissimilarity matrix was calculated by comparing each image within the category to each other image within that category. After using methods described above to generate a dissimilarity matrix for each image to all images within an object category, putative pairs were created using the two images with the lowest dissimilarity score. Pairs were sampled without replacement from the matrix until all images were paired, yielding $>400,000$ potential pairs at a range of dissimilarities. In the case of MST lure images, original image pairings were maintained, but dissimilarities were calculated like other image pairs. Selected pairs were spread evenly by dissimilarity across session while making sure that each object category was presented as lure pair no more than four times across an entire session.

After lure pairs were selected, extra images from unused pairs were used to populate each experimental session as exactly repeated images. These images were never shown in the same session as its potential pair.

Neural recording

The activity of units in HC and ITC were recorded in two monkeys while they performed the modified mnemonic similarity task. Following on

Sakon and Suzuki (2019), we targeted HC subregions CA3 and dentate gyrus. We do not have the resolution required to differentiate CA3 from dentate gyrus and report both as HC.

In Monkey 1, HC recordings were made in a single chamber in the left hemisphere, and ITC recordings were made across two chambers placed over left and right ITC. In Monkey 2, all recordings in both HC and ITC were made using a single chamber over right hemisphere. Chamber placement was guided by anatomical magnetic resonance images of the monkey's brain. In Monkey 1, chambers were centered 15.5 mm lateral of midline, and 19 mm anterior to the ear canals, with a 2° angle perpendicular to midline oriented dorsomedial to ventrolateral. HC recordings were sampled from 13 locations spanning from 13 to 16 mm anterior of ear canals, 16–21 mm lateral of midline, and depths 8–12 mm above the ventral surface of the temporal lobe in the left hemisphere. ITC recordings utilized six recording locations over both hemispheres spanning 15–18 mm anterior to ear canals, 19–23 mm lateral of midline, and of 2–5 mm above the skull (following ventral surface of the temporal lobe).

In Monkey 2, the chamber was centered at 10 mm lateral of midline to the right, and 15 mm anterior to the ear canals, with a 9° angle perpendicular to midline oriented dorsomedial to ventrolateral. Recordings in HC utilized seven recording locations spanning from 11 to 13 mm anterior of ear canals, 11–14 mm lateral of midline, and at recording depths 8–12 mm above the ventral surface of the temporal lobe in the right hemisphere. ITC recordings utilized seven recording locations in the hemispheres spanning 9–12 mm anterior to ear canals, 18–21 mm lateral of midline, and depths of 2–5 mm above the ventral surface of the temporal lobe.

Neural recordings were performed once the monkeys were fully trained [performance on novel/direct repeat images of above 80% accuracy, mnemonic performance modulated by n -back as reported previously in the lab (Meyer and Rust, 2018), and behavior was not disrupted by the introduction of lure images in practice sessions]. For Monkey 1, combined neural recordings and behavioral sessions occurred in two periods ~ 6 months apart. During each period, recordings took place 4–5 times a week and spanned 8 weeks. For Monkey 2, combined neural recordings and behavioral sessions occurred in a single 8 week period.

Neural activity was sampled using a 24-channel U-probe (Plexon) with channels spaced vertically along the shaft of the probe 100 μm apart. Continuous wideband neural signals were amplified, digitized at 30 kHz, and stored using the Grapevine Data Acquisition System (Ripple). Neural data was sorted into spikes manually offline (Plexon Offline Sorter). During sorting, between one and three candidate unit(s) were identified on each recording channel. Spike sorting was performed blind to any experimental conditions to avoid bias.

Session exclusion criterion

A recording session was included in the analysis if (1) the monkey completed a minimum session length of 1,200 trials; (2) at least one unit passed a visual screen comparing the evoked mean firing rate for each unit across trials before and after stimulus onset (between the window of -300 to 0 and 50 – 350 ms) using a t test with a threshold of $p < 0.1$. (3) Firing rates were stable over the course of a session through visual inspection; (4) the recording contained a minimum RS or RE magnitude of 2% for direct repeated images. In addition, (5) some sessions were eliminated to match memory between brain regions in the same monkey. This step was performed to roughly match on SCC predicted memory performance at the end of the image viewing window (200–500 ms). To match memory information between the two datasets, the last seven recorded sessions from monkey 1 ITC were not analyzed. For the HC populations: In Monkey 1, 26 sessions were recorded and 9 were removed (4 and 5 due to criteria 1 and 4, respectively). In Monkey 2, 17 sessions were recorded and 6 were removed (2, 2, and 2 due to criteria 1, 2, and 4, respectively). For the ITC populations: in Monkey 1, 18 sessions were recorded and 8 were removed (1, 1, 2, and 4 due to criteria 1, 2, 3, and 5, respectively). In Monkey 2, 19 sessions were recorded and 5 were removed (1, 2, and 2 due to criteria 2, 3, and 4, respectively). The sample size was chosen to approximately match that of our previous work (Meyer and Rust, 2018; Mehrpour et al., 2021).

Pseudopopulation generation

Each session was limited to 24 channels and yielded too little data for an assessment of alignment between neural and behavioral performance. To overcome this challenge, we combined data across sessions into larger pseudopopulations using methodology introduced by Meyer and Rust (2018). When creating these pseudopopulations, we took trials at our target n -back (~ 24 for Monkey 1 and ~ 8 for Monkey 2) and aligned data across sessions, splitting the data into four conditions: (1) novel images in the “exact repeat” condition, (2) novel images in the “lure” condition, (3) repeated images in the “exact repeat” condition, and (4) lure images. Lure pseudopopulations were aligned by CORnet_S-IT dissimilarity. For novel/exact repeat images, the pseudopopulation images were randomly aligned, consistent with dissimilarity values of 0.

Because different images were used in each session, aligning images in this way implicitly assumes that (1) the total number of spikes evoked in a brain area is matched across different images, (2) the data recorded in any one session is a representative sample of those statistics, and (3) the responses of the units recorded in different sessions are uncorrelated.

The number of trials in each pseudopopulation was limited to the length of the shortest session, meaning that each session that was longer than the shortest had trials ablated until it matched in size. In order to trim populations, images to be ablated were chosen randomly. This process yielded four pseudopopulations for each brain region for each monkey: two lure pseudopopulations of the responses to each of the lure image pairs, and two exact repeat pseudopopulations of the novel and repeated response to the image.

Finally, the number of units included in each population was matched for memory information using SCC classifier performance in the viewing window 200–500 ms. This resulted in the following performances (mean $\pm$ SE): Monkey 1—HC: 0.850 ± 0.00013 , ITC: 0.865 ± 0.00011 ; Monkey 2—HC: 0.758 ± 0.00018 , ITC: 0.744 ± 0.00016 . For Monkey 1, pseudopopulations consisted of 422 HC and 504 ITC units, each exposed to 213 repeated and 133 lure images. For Monkey 2, 340 HC and 595 ITC units were used, with 142 repeated and 80 lure images.

The analyses in Figure 6 used pseudopopulations created by combining data from both monkeys, meaning pseudopopulations were limited to the smaller of the two paired populations resulting in sizes. For the combined analysis, pseudopopulations included 747 HC units and 1,024 IT units, each exposed to 142 repeated and 80 lure images.

d'

Unit d' was calculated as the difference between a unit's mean response to novel images (between 300 and 500 ms) compared with the response to the same image when it was repeated, divided by the average standard deviation across both sets (Fig. 3*b,c*).

Total spike count and weighted linear decoders

Following methods described in Mehrpour et al. (2021), pseudopopulation responses were decoded from HC and ITC using two linear decoding schemes. For both decoders, the population response was quantified as the vector $\mathbf{x}$ containing spike counts on a given trial. To ensure that the decoder did not erroneously rely on visual selectivity, the decoder was trained on balanced pairs of novel/repeated trials in which monkeys viewed the same image (regardless of behavioral outcome or experimental condition).

Cross-validated training and testing. We applied the same, iterative cross-validated procedure for each linear decoder. On each iteration of the resampling procedure, the responses for each unit were randomly shuffled within each experimental condition to ensure that artificial correlations (e.g., between units recorded within a sessions) were removed. For exact repeat images, this shuffling was done across the entire pseudopopulation. Lure images were separated into three bins based on ranked similarity before shuffling, resulting in three bins of different dissimilarity. Each iteration was cross-validated by reserving a subset of direct repeat images equal to the number of images in each lure bin (46 for Monkey 1, 26 for Monkey 2, and 26 for the combined population; roughly 25% of the novel/repeat responses). The remaining direct repeat and novel trials were used to train one of the linear decoders to distinguish novel versus repeated images. A criterion was then generated serving as a threshold to classify the response to a given image as a novel or

repeat. This threshold was then used to predict whether the withheld novel/repeat images, as well as the lure images, were novel or repeated. A neural prediction of the proportion of trials on which “repeated” would be reported was computed as the proportion of test trials in a given condition that took on a value less than a criterion. Finally, the predicted response pattern was rescaled by a rescaling parameter as a proxy for adjusting the population size to match performance. This rescaling factor was computed by adjusting direct novel and repeat performance to 100% accuracy. This rescaling factor was then applied to lure performance.

Specifically, all decoders in this study took the general form of linear discriminators. The class (novel/repeated) of a population response vector $\mathbf{x}$ was determined by the sign of the following:

$$f(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} - b, \quad (1)$$

where $\mathbf{w}$ is an N -dimensional weight vector in the N -dimensional IT neural space (N is the number of units) and b is decision criterion, given by the following:

$$b = \frac{1}{2} \mathbf{w} \cdot (\boldsymbol{\mu}_N + \boldsymbol{\mu}_R), \quad (2)$$

Here $\boldsymbol{\mu}_N$ and $\boldsymbol{\mu}_R$ are the mean population response vectors across novel and repeated images in the training set, respectively. A population response vector $\mathbf{x}$ was classified as “novel” if $f(\mathbf{x}) > 0$ and “repeated” if $f(\mathbf{x}) < 0$.

The total spike count classifier (associated with RS) used a homogeneous weight vector (Figs. 5, 6):

$$\mathbf{w}_{RS} = \mathbf{1} = (1, 1, \dots, 1), \quad (3)$$

The weighted linear decoding vector (which could, in principle, combine a population of heterogeneous units in which some exhibit RS and others RE) was defined as follows (Fig. 4):

$$\mathbf{w}_{WD} = (\boldsymbol{\mu}_N - \boldsymbol{\mu}_R), \quad (4)$$

where $\boldsymbol{\mu}_N$ and $\boldsymbol{\mu}_R$ are the mean population response vectors across novel and familiar images in the training set, respectively. This is a simple form of a fisher linear discriminant that arises when the average covariance is a multiple of the identity and is sometimes called a “prototype classifier.”

Adding ranked units. Assessing the contributions of units to memory performance as a function of their ranked d' -prime was analyzed by first generating the d' -prime for each unit within a population based on the training data and then adding units from a pool. This pool of units was tested for cross-validated performance as described above. For each run of the analysis, a ranking type (i.e., ranking by positive weights amplitude “High”; negative d' -prime amplitude “Low”; or a random/shuffled ranking “Rand”) dictated which units were added. For each combination, classifier performance was gathered for each 10% of the total population (i.e., 10, 20, 30... 100%; Fig. 4).

Sharpening metric

The goal of the sharpening metric was to quantify the degree of sharpening present in either a behavioral choice pattern or a prediction based on a neural classifier. This sharpening was quantified relative to a benchmark connecting the hit rate (repeated correct performance) to the false alarm rate (novel incorrect performance) by integrating the area of the curve with respect to this benchmark. To compare classifier sharpening and behavior, both classifier predictions and behavior were rescaled to 100% for repeats and novel images, and lure accuracy was rescaled proportionally. Specifically, this rescaling took classifier performance (specifically, average percent chose repeat) C to all images and rescaled it using classifier performance to novel C_N and repeat C_R images as bounds to yield a rescaled percent chose repeat for all image dissimilarities C_{Rescaled} :

$$C_{\text{Rescaled}} = (C - C_R) / (C_N - C_R), \quad (5)$$

Next, the area below the curve relative to the linear benchmark was added, and area above the benchmark was subtracted. Under the assumption that lure performance is intermediate to novel and repeat performance, this value ranges from -1 to 1 . Finally, to capture signal rather than noise, the result was weighted by the squared average performance of the classifier to novel and repeat images ranging from a weighting of 1 (at the highest observed classifier performance during the viewing period) to zero (when the classifier performance was at chance). The resulting sharpening metric ranged from -1 to 1 , with positive scores representing sharpening, negative scores representing broadening, and 0 reflecting no change relative to the ITC benchmark (Fig. 6).

Experimental design and statistical tests

Behavioral and neural data were collected from two adult rhesus macaque monkeys (one female) over 52 behavioral sessions (27 and 25 from Monkeys 1 and 2, respectively). For each session, a minimum of 213 repeated pair and 133 lure pair image responses were recorded from Monkey 1 and 142 repeated pair and 80 lure pair image responses from Monkey 2. Recorded neural data included 422 HC and 504 ITC units from Monkey 1, and 340 HC and 595 ITC units from Monkey 2 (see above, Neural recording, for details on recording locations). Recorded sample size of both the number of subjects used and the number of units recorded were drawn from previous studies (Meyer and Rust, 2018; Jaegle et al., 2019; Mehrpour et al., 2021). Final population sizes of neural and behavioral data were determined using criteria described above in “pseudopopulation generation.”

To determine 95% confidence intervals for the behavior in Figures 2d and 5 and Figure S1, we computed the mean performance across sessions and computed the interval as the mean $\pm 1.96 \times$ standard error of the mean.

To assess d' distributions with respect to zero for Figure 3, a one-sample two-way t test was used. To assess if the distribution of sharpening values fell below the benchmark for Figure 5, a one-sample, one-way t test was used. To assess temporal differences between brain regions in Figure 6, a sharpening value was generated for each of 21 time bins on each classifier iteration based on the population response, and latency was defined as the time at which a sharpening threshold of 0.1 was first crossed. For a small fraction of iterations, noise triggered an impossible latency before stimulus onset, and these were eliminated from further analysis. Distributions of latencies were then compared across brain areas using a two-sample two-way t test. All statistical tests were computed using Matlab 2024b.

Data accessibility

Data and code available at <https://doi.org/10.6084/m9.figshare.30813176>.

Results

The adapted mnemonic similarity task

The monkeys' behavioral task was an adaptation of the mnemonic similarity task (MST) that has been used extensively to measure the contribution of HC to shaping the fidelity of visual memory (Kirwan et al., 2007; Bakker et al., 2008). In the original task, subjects viewed images that were novel, exactly repeated, or similar to those viewed before, where degree of similarity was determined empirically as the likelihood that a lure image was mistakenly classified as repeated, on average, across many subjects. Here, we take a complementary approach where we measure similarity by benchmarking against representational dissimilarity based on a state-of-the-art model of ITC: CORnet_S-IT (Kubilius et al., 2019; see Materials and Methods). This network has been shown to be predictive of neural data from macaques observing images and behavioral predictions on object identification tasks from human participants (Kubilius et al., 2019). The rationale behind relying on a model of ITC for these experiments is twofold. First, our single-exposure memory experiments require exceedingly large numbers of lure pair images ($\sim 30,000$) at a range of visual

dissimilarities, which is a prohibitively large number to test and curate empirically. Second, benchmarking against ITC representations allows us to assess from behavior alone, the extent, if at all, that nonlinear sharpening computations happen downstream of ITC visual representations.

In this task, monkeys viewed images and reported whether they were novel (i.e., never before seen in training or testing) or repeated (meaning that they had seen that image exactly once before and earlier in the experimental session; Fig. 2a,b). Monkeys initiated each trial by fixating a point at the center of the screen, followed by a brief delay and then the presentation of an image (Fig. 2a). After 500 ms of fixating on the image, a go cue appeared, indicating that the monkeys were free to make their selection via a saccade to one of two response targets. Correct responses were rewarded with juice. Lure images were similar but not identical to novel images and contained an object from the same category, systematically titrated at a range of visual similarities (Fig. 2d, bottom). Only a small fraction (13%) of images were lures.

The first presentation of a lure image was rewarded when the monkey chose “novel.” Lure images were also repeated, and on those trials, they were rewarded when the monkey chose “repeat.” (Nonlure) repeated images, lure images, and lure repeats were shown $\sim N$ trials following their lure pair, where N was adjusted to equate the performance across the two monkeys (~ 24 trials for Monkey 1 and ~ 8 trials for Monkey 2). Novel images that were paired with lures were never repeated. To minimize overall bias in the experiment (for more novel as compared with repeated images), lure images were repeated. Typically, a single-exposure memory experiment would aim to balance novel and repeated exposures so that each image is shown exactly twice and half the correct answers are “novel” and half are “repeated.” However, the introduction of lure images (rewarded as novel) shifted the distribution of correct responses. To reduce the impact of this, lure images were repeated. Consequently, while the first image presented in each session was always novel, the probability of subsequent images being novel versus repeated quickly converged to 57%. The slight increase in novel images is due to novel images that are paired with lures and not repeated again. In each behavioral session, monkeys viewed 1,500 images consisting of novel, exact repeats, lures, and lure repeats.

Performance was measured as the percent chose repeat (PCR) for any given trial category (Fig. 2c,d). For novel images, this represents the false alarm rate, and for repeated images, this represents the hit rate. For lure images, the PCR is the rate of false memory or equivalently the mnemonic confusability of the lure images. While our experimental design has a modest bias toward novel answers (57% of trials), the monkeys were largely unbiased in their responses [for direct repeats, M1: false alarms (reporting novel as familiar) = 1.3 ± 0.6 , misses (reporting familiar as novel) = 2.5 ± 1.1 ; M2: false alarms = 5.7 ± 0.9 ; misses = 8.9 ± 1.6].

Assessing hypotheses about “sharpening” (or broadening) downstream of ITC visual representations requires specifying a benchmark of “no sharpening” to measure against. Here we use a method inspired by representational similarity analysis (Kriegeskorte et al., 2008) where we measured dissimilarity based on ITC visual representations. Specifically, we used 1 minus the Pearson's correlation of the ITC population responses on two trials, where the ITC responses are either estimated by an ITC model or are measured directly or indirectly from brain activity (Kriegeskorte et al., 2008; see Materials and Methods). This can be thought of as plotting visual representations on the x -axis and memory representations on the y -axis. If visual memory

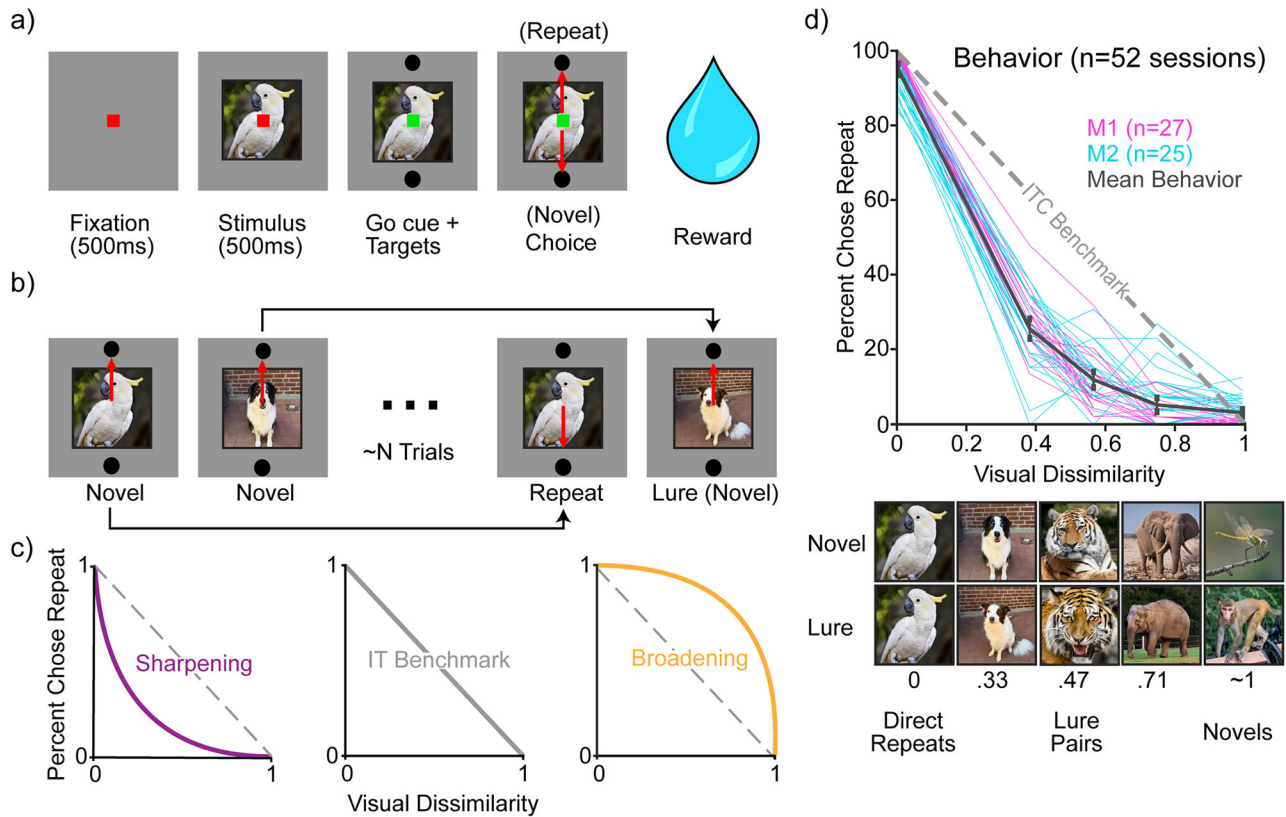


Figure 2. Adapted mnemonic similarity task (MST). *a*, Each trial began with the monkey fixating a red dot for 500 ms. If fixation was successfully maintained, an image was presented for 500 ms before the fixation square turned green and targets appeared above and below the image. The monkey was then free to make a saccade to one of the two targets to report whether they judged the image to be novel or repeated. Successful choices were rewarded with juice. *b*, Task overview: An example sequence of trials which include novel, exactly repeated, and lure images were separated by N trials where N was adjusted to equate performance between the monkeys ($N = \sim 24$ trials in Monkey 1 and ~ 8 trials in Monkey 2). Black arrows connect novel/repeat and novel/lure pairs. *c*, Hypothetical behavioral patterns illustrating different possible outcomes. The diagonal gray dashed line (middle) represents the expected performance of the monkey if they performed exactly as well as would be expected based on our measure of ITC dissimilarity. Shown are hypothetical examples of behavioral sharpening (left) and broadening (right) relative to this benchmark. *d*, Behavioral performance. Top, Monkey 1 and Monkey 2 sessions plotted in magenta and cyan, respectively. Mean behavioral performance across all 52 sessions is plotted in black with 95% confidence interval error bars. Behavioral reports of the monkeys indicate evidence of sharpening on average and within individual sessions. Bottom, Example lure image pairs at a range of visual dissimilarities. Dissimilarity values were quantified by the similarity of lure pairs based on a model of ITC: CORnet_S-IT (see Materials and Methods).

behavior aligns with this benchmark, it indicates that the transformation from ITC visual representations to visual memory behavior is linear. Conversely, deviations from this benchmark indicate that this transformation is shaped by nonlinear computations (which may include hippocampal pattern separation).

We found clear and robust evidence for behavioral sharpening for both monkeys, including within individual sessions (Fig. 2*d*, magenta and cyan lines). This sharpening could follow in whole or in part from hippocampal pattern separation computations, or it could follow from additional computations at one or more earlier stages in the processing hierarchy (which includes computations within ITC itself, as well as perirhinal cortex, parahippocampal cortex, and entorhinal cortex; Felleman and Van Essen, 1991). To investigate, we compared sharpening behavior with memory representations at the ends of this processing chain: in HC and ITC.

Neural analogs of sharpening in HC and ITC

As the monkeys performed the task, we recorded neural spiking responses from HC or ITC using a multichannel electrode acutely lowered before the monkey began each session. For quality control, each recording session was screened based on its recording stability across the session, the number of behavioral trials completed, visual responsiveness, and memory information

(see Materials and Methods). Data reported here correspond to the subset of images that have a valid behavioral response—meaning the monkey completed the trial without breaking fixation—for both the first and second presentations of the image (either a novel/exact repeated or a novel/lure pair).

Comparisons between neural responses and behavior in single-trial memory tasks like this require many hundreds of units (Meyer and Rust, 2018). To analyze the neural data, we concatenated units across sessions to create larger pseudopopulations. When creating these pseudopopulations, we aligned novel, repeated, and lure images based on their lure dissimilarity (see Materials and Methods). The resulting pseudopopulations consisted of the responses to novel and repeated images, as well as pairs of novel and lure images. These populations were matched for their numbers of image pairs across brain regions within each monkey and condition (resulting in 213 and 142 novel/repeated image pairs and 133 and 80 novel/lure image pairs in Monkey 1 and Monkey 2, respectively). Within each monkey and across the two brain areas, the number of units included in each population was matched for memory information using classifier performance (see Materials and Methods). The resulting dataset included $n = 422$ HC and $n = 504$ ITC units for Monkey 1 and $n = 340$ HC units and $n = 595$ ITC units for Monkey 2.

Having finalized the neural populations, we next sought to relate them directly to behavior. A decoding scheme is required to relate neural representations of memory with memory behavior. When memory is reflected as RS, that decoding scheme can be approximated by a simple total spike count linear decoder or “spike count classifier” (SCC); however, when memory is reflected as mixtures of RS in some units and repetition enhancement (RE) in others, that linear decoder requires mixtures of positive and negative weights. Following on earlier work suggesting that visual memory information is reflected in ITC as RS, we hypothesized that this is true in HC as well.

An examination of the grand mean firing rate responses to novel and repeated images (Fig. 3*a*) revealed that RS increased in strength over the 500 ms. We thus chose to evaluate memory performance by counting spikes in a window 300–500 ms following stimulus onset (and unless stated otherwise, this holds for all analyses presented). For each unit, d' was calculated by comparing the novel and repeated responses (see Materials and Methods). We found that the distributions of d' were roughly normal but shifted significantly away from zero in both HC and ITC (HC: mean = 0.095, $t_{(746)} = 18.86$, $p = 3.40 \times 10^{-65}$, two-tailed one-sample t test; ITC: mean = 0.082, $t_{(1,023)} = 21.93$, $p = 1.04 \times 10^{-87}$, two-tailed one-sample t test), indicating populations dominated by RS along with some noise (Fig. 3*b,c*).

To determine whether any of the negative d' units reflected reliable memory information as RE (as opposed to noise), we compared cross-validated classifier performance for a weighted linear decoder that can take the sign of each unit into account by assigning RE and RS units opposite signs. Specifically, we computed memory classification performance as we added units ranked by their signed classifier weights as high-to-low (red), low-to-high (blue), or randomly selected (black), based on training data alone (Fig. 4). In these plots, above-chance performance for small numbers of units with low-to-high ranking would be indicative of RE contributing to classifier performance, whereas above-chance performance for small numbers of units with high-to-low ranking is indicative of RS contributing to classifier performance. We found that trends in performance were dominated by the latter pattern, with negligible evidence for the former in HC and ITC of both monkeys (Fig. 4). Together, these

results suggest that memory information in both HC and ITC is reflected as RS (as opposed to mixtures of RS in some units and RE in others). Below, we compare neural memory representations with behavior using a total spike count linear classifier (which decodes RS).

To compare neural representations with behavior, the spike count decoder was trained on novel and repeated images, and performance was assessed both with cross-validated images from the same conditions as well as for its ability to generalize to lure conditions. Qualitatively, like behavioral sharpening, classifier predictions based on HC neural responses reflected sharpening in both animals (Fig. 5*a,b*, purple). More surprising was that classifier predictions based on ITC neural responses also clearly reflected robust sharpening relative to the benchmark of ITC visual representations in both monkeys (Fig. 5*c,d*, purple). We quantified sharpening based on a metric that computed area of the neural classifier prediction curve relative to the ITC benchmark, weighted by overall classifier performance (see Materials and Methods). With this metric, positive values reflect sharpening whereas negative values reflect broadening (Fig. 2*c*). Classifier predictions in both animals and both brain areas were significantly sharpened relative to the ITC benchmark of zero sharpening (Fig. 5*a–d*; HC Monkey 1: mean = 0.154, $t_{(99)} = 27.60$, $p = 2.74 \times 10^{-48}$; HC Monkey 2: mean = 0.267, $t_{(99)} = 30.38$, $p = 5.54 \times 10^{-52}$; ITC Monkey 1: mean = 0.207, $t_{(99)} = 59.03$, $p = 1.40 \times 10^{-57}$; ITC Monkey 2: mean = 0.242, $t_{(99)} = 19.81$, $p = 3.17 \times 10^{-36}$; one-tailed one-sample t test). Importantly, that ITC neural predictions reflected sharpening (rather than the no sharpening benchmark predicted by ITC visual representations) suggests the existence of a nonlinear transformation between ITC visual and memory representations.

To validate that these results were not due to individual units with large sporadic firing rates, we reran the analysis after normalizing firing rates for each unit by z -scoring the spike firing rate responses across all images. This manipulation did not qualitatively change the results (Fig. S1). As expected from the nature of spike count decoder, classifier performance aligned with the distributions of spike firing rates for each of the lure distributions (Fig. S2).

Finally, we wondered whether HC and ITC visual memory representations emerged relative to stimulus onset as sharpened (as

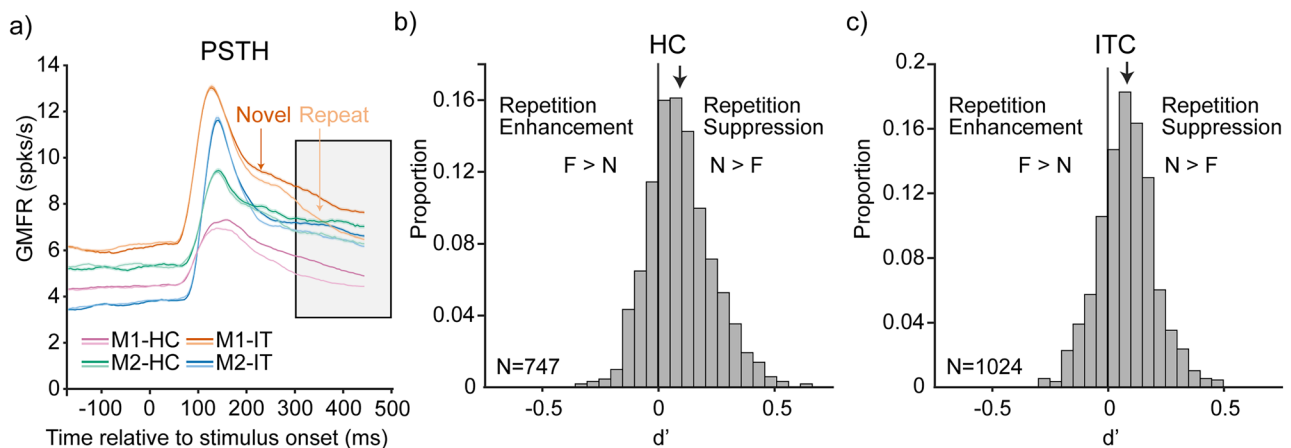


Figure 3. Memory information is reflected as RS in both HC and ITC. *a*, Grand mean peristimulus time histogram (PSTH) of neural responses to novel and repeated images, averaged across all units in each population and response type. Traces smoothed by averaging over a 50 ms time bin. Monkey1-HC, Monkey2-HC, Monkey1-IT, and Monkey2-IT are plotted as magenta, green, orange, and blue, respectively. Novel images are plotted as darker lines and repeated images are plotted as lighter lines. Standard error of the means plotted as tight areas around lines. Gray shaded box (300–500 ms) indicates the time window over which we computed d' . *b*, *c*, Histograms of d' values for individual units in HC and ITC. Arrows indicate population means (HC = 0.095; ITC = 0.082). Values greater than zero have positive d' , indicating RS. Values less than zero have negative d' , indicating RE.

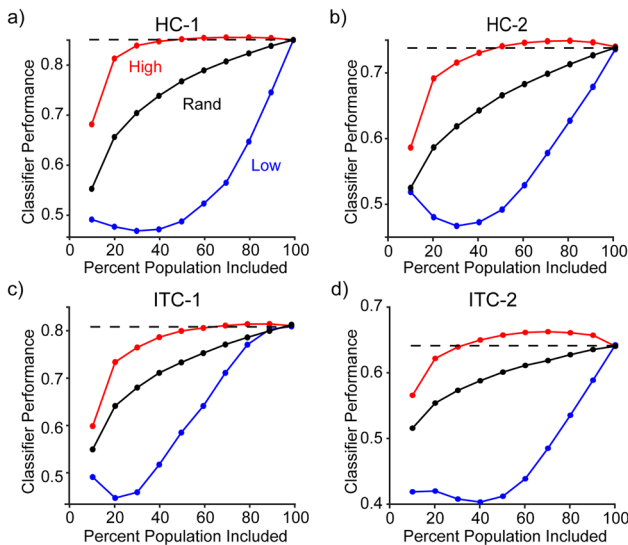


Figure 4. Population-based confirmation that memory information is reflected in HC and ITC as repetition suppression. **a–d**, Performance of a weighted linear classifier as a function of population size with respect to units ranked by classifier weight. Each plot shows performance as units are added to the population that are ranked by signed classifier weights high-to-low (“High”-red; where first-ranks are RS), low-to-high (“Low”-blue; where first-ranks are RE), or randomly (“Rand”-black). Full-sized population classifier performance is reflected by the dashed horizontal line. **a, b**, Hippocampus populations for Monkey 1 (left) and Monkey 2 (right). **c, d**, The same, but for ITC populations.

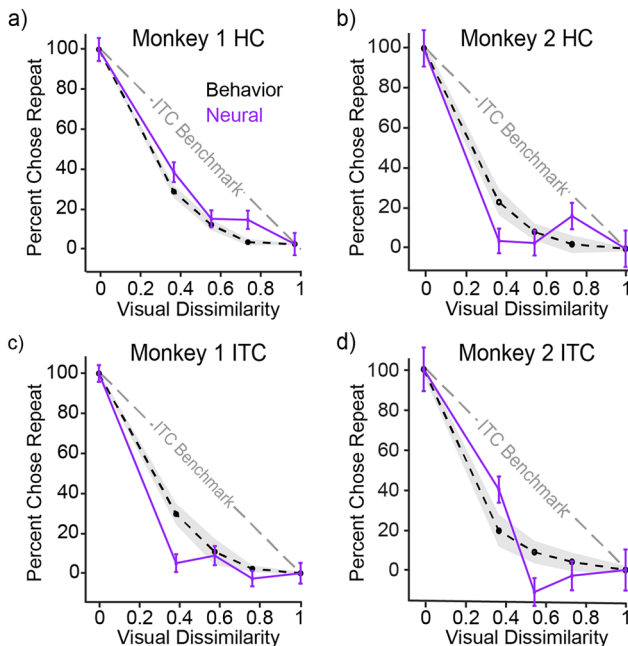


Figure 5. Behavioral sharpening is reflected in both HC and ITC neural representations of memory. **a–d**, The comparison of sharpening behavior (black, dashed), cross-validated spike count classifier performance (purple), and ITC benchmark (gray). The black dashed line shows the mean behavioral performance across all sessions with 95% confidence intervals shown with a gray shaded region. The purple line reflects mean and standard error computed from cross-validated classifier predictions of behavior, where classifier predictions have been rescaled such that novel and repeated performance is matched to behavior and lure prediction is scaled proportionally (see Materials and Methods).

opposed to linearly with sharpening later) and whether there were any obvious differences in the dynamics of how they emerged in HC versus ITC. For these analyses, we combined the data from

the two monkeys into one population to maximize SNR and the sensitivity to detect differences. Consistent with the intuitions presented in Figure 5, this metric reflected sharpening in both HC and ITC at the end of the viewing window that approximately aligned with behavior (Fig. 6a, gray arrow at 400 ms, reflecting a spike count window 300–500 ms). Likewise, visual memory representations in HC and ITC emerged with approximately the same dynamics (Fig. 6a, arrows at 150, 275, 350, and 400 ms; Fig. 6b). To statistically compare the dynamics of sharpening in HC versus ITC, the latency of each was quantified as the time at which cross-validated classifier performance first crossed a sharpening threshold of 0.1, across classifier iterations. The resulting latency distributions were not significantly different (HC mean = 260.6 ms, ITC mean = 251.6 ms, $t_{(312.71)} = 1.92$, $p = 0.055$; two-sample t test). These results reveal no clear evidence of sharpening emerging first in HC, followed by ITC, which could have been consistent with sharpening in HC that was fed back to ITC. If anything, our results suggest a (nonsignificant) trend in the opposite direction.

All of these results rely on comparison with the benchmark of ITC visual representations generated using CORnet_S-IT as a model (Kubilius et al., 2019). However, there are limitations of how effective artificial neural networks serve as models for brain region representations (Schaeffer et al., 2022; Ahlert et al., 2024; Feghhi et al., 2024) and of particular concern for this experiment are model inaccuracies that nonlinearly reflect actual ITC representational similarity, as these could spuriously lead to the conclusion that visual memory representations are sharpened. Indeed, if a nonlinear relationship exists between CORnet_S-IT dissimilarity and ground-truth neural dissimilarity, the sharpening we have observed (Figs. 2, 5, 6) could be an artifact. While our experiment was not optimized to estimate neural similarity (as each image was presented at most twice), the data do have some power to make these estimates. To test for nonlinearities, we thus compared CORnet_S-IT and neural estimates of dissimilarity for novel–lure pairs at different time points following stimulus onset. At all times, these were well-described as linear with no indication of a nonlinear relationship (Fig. S3). Similarly, we performed simulations to determine how noise might impact the assumptions underlying our ITC benchmark as well as our interpretation of the data. These simulations revealed that the impact of noise is linearization (not sharpening), and the behavioral and neural responses we observed can be accounted for by nonlinearly sharpened RS plus noise (Fig. S4).

Discussion

This study aimed to test the nature of the transformations from ITC visual representations to HC visual memory representations. Based on prior work (Yassa and Stark, 2011; Bennett et al., 2019; Sakon and Suzuki, 2019), we expected to find that HC visual memory representations mimicked behavioral sharpening, and indeed, we found evidence for this in both monkeys. However, an open question is the extent that hippocampal pattern separation was primarily responsible for this nonlinear transformation or if other brain regions such as ITC also reflect some degree of sharpening—surprisingly, we found this to clearly be the case.

Having found robust evidence for sharpening in ITC visual memory representations, reflected as RS, one question is: What might the source of sharpened ITC RS be? Here, there are four possibilities to consider. The first two relate to processing before or within ITC. First, ITC RS might be the result of cumulative RS computations in the ventral visual pathway prior to ITC (from structures such as V2 and V4; Williams and Olson, 2022).

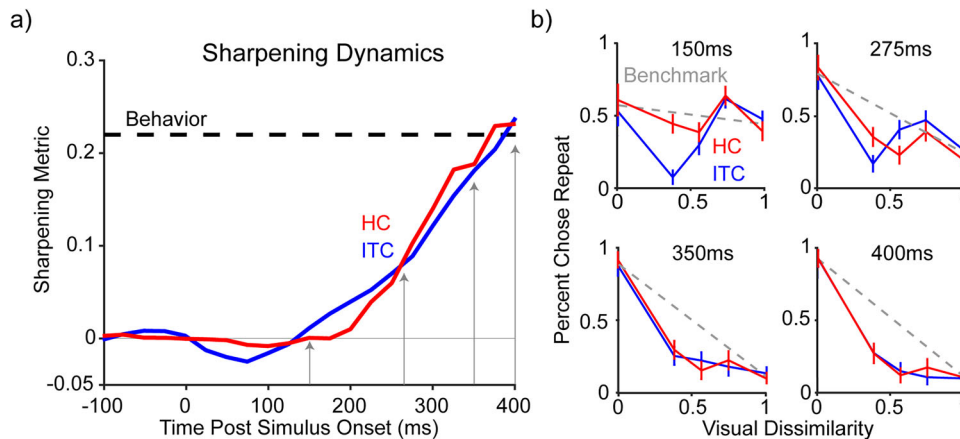


Figure 6. Neural sharpening emerges in both HC and ITC. **a, b**, Dynamics analyses performed using a 200 ms window on a combined population of both monkeys across HC and ITC (red and blue, respectively). **a**, Dynamics of sharpened classifier performance (see Materials and Methods for sharpening metric). Behavioral sharpening plotted as a black dashed line. Gray arrows indicate centers of time slices shown in panel **b**. **b**, Individual 200 ms time slices of classifier accuracy as a function of dissimilarity. The dashed gray line connects average HC and IT performance for novel and repeated images, or equivalently, the ITC benchmark. To illustrate classifier performance to novel and repeat images during each slice, classifier performances (and thus, the ITC benchmark) have not been rescaled here. Error bars indicate standard error of the mean over classifier cross-validation iterations.

However, RS in the V2 input to ITC is weaker than within ITC itself and thus cannot alone account for it (Williams and Olson, 2022). A second possibility is that ITC RS is the result of nonlinear computations within ITC itself (Brown et al., 1987). RS is inherently an adaptation-like phenomenon that extends over long timescales, and its alignment with familiarity behavior is captured with models in which familiarity memory happens via feedforward, anti-Hebbian plasticity in an IT-like network layer (Földiák, 1990; Tyulmankov et al., 2022). Likewise, the dynamics of RS are captured by models in which familiarity memories are stored as changes in ITC synaptic weights, and RS emerges through recurrent interactions between excitatory and inhibitory neurons (Lim et al., 2015; Agnes and Vogels, 2024). In this proposal, sharpened RS emerges as part of this ITC computation.

The third possibility is that sharpened ITC visual memory representations reflect computations in the medial temporal lobe that are then fed back onto ITC. This could be the result of RS computations in structures such as perirhinal cortex (Bogacz and Brown, 2003), where evidence of feedback to ITC exists for other forms of memory (paired associations; Miyashita, 2019). Likewise, memory representations in ITC could reflect computations that happen in HC that fed back to cortex, analogous to how other forms of learning are thought to occur (Kim et al., 2023). A final possibility is that sharpened ITC visual memory representations follow from computations that rely on reciprocal interactions between multiple brain areas that include ITC as well as one or more structures such as perirhinal cortex, entorhinal cortex, HC, and/or prefrontal cortex (Ranganath, 2006; Newmark et al., 2013; Li et al., 2015; Sućević and Schapiro, 2023). From our data alone, we are unable to fully clarify the contributions from each class of model toward sharpened RS in ITC. However, a parsimonious explanation of our results is that sharpened ITC RS is not an epiphenomenon but indicative of a contribution of ITC to shaping the fidelity of visual memory. Future experiments will be required to tease apart exactly what that contribution is.

A different possibility is that what we are reporting as sharpening does not follow from computations downstream from ITC visual representations, but transformations of ITC visual representations over time that are then stored in memory. The model of ITC we relied on has been compared most extensively with the early

stimulus-evoked response (<200 ms following stimulus onset), and less is known about how it aligns with ITC visual representations at later times. To explore whether ITC visual representations sharpen over time, we compared ITC model predictions with the ITC visually evoked response in later time windows, and we found no evidence for it (Fig. S4). Even if the hypothetical scenario were responsible, it would still mean that sharpening computations in ITC downstream of the early visually evoked response would contribute to shaping the fidelity of visual memory behavior.

A significant body of research in humans suggests that an intact and functioning HC is required to support the fidelity of visual memory (Kirwan et al., 2007; Bakker et al., 2008; Wixted and Squire, 2010; Lacy et al., 2011; Yassa and Stark, 2011; Stark et al., 2013, 2019; Suthana et al., 2015; Barbeau et al., 2017; Bennett et al., 2019). Namely, individuals with impairments to HC are more likely to mistakenly report that they have seen visually similar lures than control subjects. Likewise, evidence from human fMRI (Nash et al., 2021) and single unit recordings (Suthana et al., 2015) suggest that visual memory representations are more sharpened in HC than its input, entorhinal cortex. One mystery of our results is why we fail to qualitatively replicate this result, as visual memory representations were no less sharpened in ITC than HC in our data. One possible explanation is a species difference—although evidence suggests that “pattern separation” computations happen even in rodents in the context of spatial recognition tasks (van Goethem et al., 2018) and that rodent HC is required for innate object recognition memory behavior (Broadbent et al., 2010; Cohen et al., 2013; Cohen and Stackman, 2015; Johnson et al., 2017). Another possible explanation is that HC may contribute to sharpening when conceptual (rather than just perceptual) factors are at play (Barense et al., 2007; Bonnen et al., 2021; Gurguryan et al., 2024), acknowledging that the monkeys in this study were largely unfamiliar with the objects presented in the images used in these experiments. Of note is that neither of these possible scenarios undermines the conclusion that ITC contributes in some way to shaping the fidelity of visual memory.

We observed a majority of visually responsive units that were recorded in ITC and HC showed RS, with only a few showed RE. Furthermore, RS units carried classifier predictive power and were matched to behavior. This is reminiscent of previous

work from Sakon and Suzuki (2019), who reported that visual memory representations in HC were reflected as mixtures of RS in some units and RE in others; however, the units that reflected the most RS were able to discriminate between repeat and lure images whereas RE units were not. Of note is that the behavioral paradigm employed by Sakon and Suzuki (2019) centered on preferential looking (where monkeys innately prefer looking at novel images longer than repeated ones), thus combining a motivational component (curiosity) with familiarity. Together, these studies suggest that, like ITC, HC visual recognition memory representations are reflected as RS whereas RE may reflect other processes in these brain areas (see also Kim, 2017). While we did not find differences in the patterns of RS between ITC and HC (e.g., as a function of visual dissimilarity), the single-trial nature of our data (imposed by the single-exposure nature of visual familiarity behavior) may limit our ability to detect such differences.

Other work has reported a different sharpening phenomenon with familiarity. Namely, following extensive exposure to the same images, the visual selectivity of excitatory ITC neurons becomes more sparse (Woloszyn and Sheinberg, 2012; Lim et al., 2015), and the stimulus-evoked response becomes shorter in duration (Meyer et al., 2014). It is unclear how that phenomenon relates to our results, as our version of sharpening focuses not on visual selectivity for the same images but rather the nature of memory transfer across different but similar images. Conceptually, the two could, in principle, be related insofar as viewing a novel image could sharpen the ITC population response, thereby narrowing the degree to which memory information transfers across images. However, the timescale of the two phenomena differs in that repetition suppression following a single image presentation lasts for minutes (Meyer and Rust, 2018) whereas changes in ITC with extensive familiarity result in longer lasting changes (days to weeks and beyond; Woloszyn and Sheinberg, 2012; Koyano et al., 2023; Mohan et al., 2024). Future work will thus be required to determine whether and how these phenomena relate.

A number of recognition memory studies have examined reaction time patterns to make inferences about underlying mechanism (Murdoch, 1985; Yonelinas and Jacoby, 1994; Weidemann and Kahana, 2016). In our experiments, patterns of reaction times were not replicated across monkeys (Fig. S5), and we are thus unable to leverage them to make similar inferences. The lack of replication for reaction time patterns may follow from our experimental design, which was not optimized to measure reaction times (to minimize other confounds).

The results we present in this paper challenge the traditional view that the hippocampus uniquely shapes the fidelity of visual memory. They suggest that instead, the fidelity of visual memory follows from integrated computations across brain regions. Understanding memory failures (from diseases to typical errors) requires an understanding of visual memory that appreciates the specific and combined contributions of neocortical regions and the medial temporal lobe. This study helps build toward a holistic model of visual memory by demonstrating that computations traditionally thought to happen only in the hippocampus are reflected in visual cortex.

References

- Agnes EJ, Vogels TP (2024) Co-dependent excitatory and inhibitory plasticity accounts for quick, stable and long-lasting memories in biological networks. *Nat Neurosci* 27:964–974.
- Ahlert J, Klein T, Wichmann F, Geirhos R (2024) How aligned are different alignment metrics? Available at: <http://arxiv.org/abs/2407.07530>
- Ally BA, Hussey EP, Ko PC, Molitor RJ (2013) Pattern separation and pattern completion in Alzheimer's disease: evidence of rapid forgetting in amnesic mild cognitive impairment. *Hippocampus* 23:1246–1258.
- Bakker A, Kirwan CB, Miller M, Stark CEL (2008) Pattern separation in the human hippocampal CA3 and dentate gyrus. *Science* 319:1640–1642.
- Barbeau EJ, Chauvel P, Moulin CJA, Regis J, Liégeois-Chauvel C (2017) Hippocampus duality: memory and novelty detection are subserved by distinct mechanisms. *Hippocampus* 27:405–416.
- Barense MD, Gaffan D, Graham KS (2007) The human medial temporal lobe processes online representations of complex objects. *Neuropsychologia* 45:2963–2974.
- Bennett IJ, Stark SM, Stark CEL (2019) Recognition memory dysfunction relates to hippocampal subfield volume: a study of cognitively normal and mildly impaired older adults. *J Gerontol B Psychol Sci Soc Sci* 74: 1132–1141.
- Berron D, Schütze H, Maass A, Cardenas-Blanco A, Kuijff HJ, Kumaran D, Düzel E (2016) Strong evidence for pattern separation in human dentate gyrus. *J Neurosci* 36:7569–7579.
- Bogacz R, Brown MW (2003) Comparison of computational models of familiarity discrimination in the perirhinal cortex. *Hippocampus* 13:494–524.
- Bonnen T, Yamins DLK, Wagner AD (2021) When the ventral visual stream is not enough: a deep learning account of medial temporal lobe involvement in perception. *Neuron* 109:2755–2766.e6.
- Brady TF, Konkle T, Alvarez GA, Oliva A (2008) Visual long-term memory has a massive storage capacity for object details. *Proc Natl Acad Sci U S A* 105:14325–14329.
- Broadbent NJ, Gaskin S, Squire LR, Clark RE (2010) Object recognition memory and the rodent hippocampus. *Learn Mem* 17:5–11.
- Brown MW, Wilson FAW, Riches IP (1987) Neuronal evidence that inferomedial temporal cortex is more important than hippocampus in certain processes underlying recognition memory. *Brain Res* 409:158–162.
- Cohen SJ, Stackman RW (2015) Assessing rodent hippocampal involvement in the novel object recognition task: a review. *Behav Brain Res* 285:105–117.
- Cohen SJ, Munchow AH, Rios LM, Zhang G, Ásgeirsdóttir HN, Stackman RW (2013) The rodent hippocampus is essential for nonspatial object memory. *Curr Biol* 23:1685–1690.
- Fahy FL, Riches IP, Brown MW (1993) Neuronal activity related to visual recognition memory: long-term memory and the encoding of recency and familiarity information in the primate anterior and medial inferior temporal and rhinal cortex. *Exp Brain Res* 96:457–472.
- Feghhi E, Hadidi N, Song B, Blank IA, Kao JC (2024) What are large language models mapping to in the brain? A case against over-reliance on brain scores. Available at: <http://arxiv.org/abs/2406.01538>
- Felleman DJ, Van Essen DC (1991) Distributed hierarchical processing in the primate cerebral cortex. *Cereb Cortex* 1:1–47.
- Földiák P (1990) Forming sparse representations by local anti-Hebbian learning. *Biol Cybern* 64:165–170.
- Goetschalckx L, Wagemans J (2019) MeMCAT: a new category-based image set quantified on memorability. *PeerJ* 7:e8169.
- Gurguryan L, Fenerci C, Ngo N, Sheldon S (2024) The neural correlates of constructing conceptual and perceptual representations of autobiographical memories. *J Cogn Neurosci* 36:1350–1373.
- Hebart MN, Dickter AH, Kidder A, Kwok WY, Corriveau A, Van Wicklin C, Baker CI (2019) THINGS: a database of 1,854 object concepts and more than 26,000 naturalistic object images. *PLoS One* 14:e0223792.
- Isola P, Xiao J, Torralba A, Oliva A (2011) What makes an image memorable? *J Vis* 11:1282.
- Jaegle A, Mehrpour V, Mohsenzadeh Y, Meyer T, Oliva A, Rust N (2019) Population response magnitude variation in inferotemporal cortex predicts image memorability (Behrens TE, Pasternak T, Buffalo EA, eds). *Elife* 8:e47596.
- Johnson SA, Turner SM, Santacrose LA, Carty KN, Shafiq L, Bizon JL, Maurer AP, Burke SN (2017) Rodent age-related impairments in discriminating perceptually similar objects parallel those observed in humans. *Hippocampus* 27:759–776.
- Khosla A, Raju AS, Torralba A, Oliva A (2015) Understanding and predicting image memorability at a large scale. In: *Proceedings of the IEEE International Conference on Computer Vision*, 2390–2398.
- Kim H (2017) Brain regions that show repetition suppression and enhancement: a meta-analysis of 137 neuroimaging experiments. *Hum Brain Mapp* 38:1894–1913.
- Kim J, Joshi A, Frank L, Ganguly K (2023) Cortical-hippocampal coupling during manifold exploration in motor cortex. *Nature* 613:103–110.

- Kirwan CB, Jones CK, Miller MI, Stark CEL (2007) High-resolution fMRI investigation of the medial temporal lobe. *Hum Brain Mapp* 28:959–966.
- Konkle T, Brady TF, Alvarez GA, Oliva A (2010) Scene memory is more detailed than you think: the role of categories in visual long-term memory. *Psychol Sci* 21:1551–1556.
- Koyano KW, Esch EM, Hong JJ, Waidmann EN, Wu H, Leopold DA (2023) Progressive neuronal plasticity in primate visual cortex during stimulus familiarization. *Sci Adv* 9:eade4648.
- Kriegeskorte N, Mur M, Bandettini P (2008) Representational similarity analysis - connecting the branches of systems neuroscience. *Front Syst Neurosci* 2:4.
- Kubilius J, et al. (2019) Brain-like object recognition with high-performing shallow recurrent ANNs. In: *Advances in Neural Information Processing Systems*.
- Lacy JW, Yassa MA, Stark SM, Muftuler LT, Stark CEL (2011) Distinct pattern separation related transfer functions in human CA3/dentate and CA1 revealed using high-resolution fMRI and variable mnemonic similarity. *Learn Mem* 18:15–18.
- Leal SL, Yassa MA (2018) Integrating new findings and examining clinical applications of pattern separation. *Nat Neurosci* 21:163–173.
- Li L, Miller EK, Desimone R (1993) The representation of stimulus familiarity in anterior inferior temporal cortex. *J Neurophysiol* 69:1918–1929.
- Li N, Chen T-W, Guo ZV, Gerfen CR, Svoboda K (2015) A motor cortex circuit for motor planning and movement. *Nature* 519:51–56.
- Lim S, McKee JL, Woloszyn L, Amit Y, Freedman DJ, Sheinberg DL, Brunel N (2015) Inferring learning rules from distributions of firing rates in cortical neurons. *Nat Neurosci* 18:1804–1810.
- Mehrner J, Spoerer CJ, Jones EC, Kriegeskorte N, Kietzmann TC (2021) An ecologically motivated image dataset for deep learning yields better models of human vision. *Proc Natl Acad Sci U S A* 118:e2011417118.
- Mehrpour V, Meyer T, Simoncelli EP, Rust NC (2021) Pinpointing the neural signatures of single-exposure visual recognition memory. *Proc Natl Acad Sci U S A* 118:e2021660118.
- Meyer T, Rust N (2018) Single-exposure visual memory judgments are reflected in inferotemporal cortex. *Elife* 7:e32259.
- Meyer T, Walker C, Cho RY, Olson CR (2014) Image familiarization sharpens response dynamics of neurons in inferotemporal cortex. *Nat Neurosci* 17:1388–1394.
- Miller EK, Desimone R (1994) Parallel neuronal mechanisms for short-term memory. *Science* 263:520–522.
- Miyashita Y (2019) Perirhinal circuits for memory processing. *Nat Rev Neurosci* 20:577–592.
- Mohan K, Pereira-Obilinovic U, Srednyak S, Amit Y, Brunel N, Freedman D (2024) Visual familiarity learning at multiple timescales in the primate inferotemporal cortex. *Neuroscience*. Available at: <http://biorxiv.org/lookup/doi/10.1101/2024.01.05.574412> [Accessed February 16, 2024].
- Monti JM, Balota DA, Warren DE, Cohen NJ (2014) Very mild Alzheimer's disease is characterized by increased sensitivity to mnemonic interference. *Neuropsychologia* 59:47–56.
- Murdock BB (1985) An analysis of the strength-latency relationship. *Mem Cognit* 13:511–521.
- Nash MI, Hodges CB, Muncy NM, Kirwan CB (2021) Pattern separation beyond the hippocampus: a high-resolution whole-brain investigation of mnemonic discrimination in healthy adults. *Hippocampus* 31:408–421.
- Newmark RE, Schon K, Ross RS, Stern CE (2013) Contributions of the hippocampal subfields and entorhinal cortex to disambiguation during working memory. *Hippocampus* 23:467–475.
- Ranganath C (2006) Working memory for visual objects: complementary roles of inferior temporal, medial temporal, and prefrontal cortex. *Neuroscience* 139:277–289.
- Sakon JJ, Suzuki WA (2019) A neural signature of pattern separation in the monkey hippocampus. *Proc Natl Acad Sci U S A* 116:9634–9643.
- Schaeffer R, Khona M, Fiete IR (2022) No free lunch from deep learning in neuroscience: a case study through models of the entorhinal-hippocampal circuit. Available at: <http://biorxiv.org/lookup/doi/10.1101/2022.08.07.503109>
- Standing L (1973) Learning 10,000 pictures. *Q J Exp Psychol* 25:207–222.
- Stark SM, Yassa MA, Lacy JW, Stark CEL (2013) A task to assess behavioral pattern separation (BPS) in humans: data from healthy aging and mild cognitive impairment. *Neuropsychologia* 51:2442–2449.
- Stark SM, Stevenson R, Wu C, Rutledge S, Stark CEL (2015) Stability of age-related deficits in the mnemonic similarity task across task variations. *Behav Neurosci* 129:257–268.
- Stark SM, Kirwan CB, Stark CE (2019) Mnemonic similarity task: a tool for assessing hippocampal integrity. *Trends Cogn Sci* 23:938.
- Sučević J, Schapiro AC (2023) A neural network model of hippocampal contributions to category learning (Zeithamova D, Frank MJ, eds). *Elife* 12:e77185.
- Suthana NA, Parikshak NN, Ekstrom AD, Ison MJ, Knowlton BJ, Bookheimer SY, Fried I (2015) Specific responses of human hippocampal neurons are associated with better memory. *Proc Natl Acad Sci U S A* 112:10503–10508.
- Tyulmankov D, Yang GR, Abbott LF (2022) Meta-learning synaptic plasticity and memory addressing for continual familiarity detection. *Neuron* 110:544–557.e8.
- van Goethem NP, van Hagen BTJ, Prickaerts J (2018) Assessing spatial pattern separation in rodents using the object pattern separation task. *Nat Protoc* 13:1763–1792.
- Vogt S, Magnussen S (2007) Long-term memory for 400 pictures on a common theme. *Exp Psychol* 54:298–303.
- Weidemann CT, Kahana MJ (2016) Assessing recognition memory using confidence ratings and response times. *R Soc Open Sci* 3:150670.
- Williams N, Olson CR (2022) Independent repetition suppression in macaque area V2 and inferotemporal cortex. *J Neurophysiol* 128:1421–1434.
- Wixted JT, Squire LR (2010) The role of the human hippocampus in familiarity-based and recollection-based recognition memory. *Behav Brain Res* 215:197–208.
- Woloszyn L, Sheinberg DL (2012) Effects of long-term visual experience on responses of distinct classes of single units in inferior temporal cortex. *Neuron* 74:193–205.
- Xiang JZ, Brown MW (1998) Differential neuronal encoding of novelty, familiarity and recency in regions of the anterior temporal lobe. *Neuropharmacology* 37:657–676.
- Yassa MA, Stark CEL (2011) Pattern separation in the hippocampus. *Trends Neurosci* 34:515–525.
- Yonelinas AP, Jacoby LL (1994) Dissociations of processes in recognition memory: effects of interference and of response speed. *Can J Exp Psychol* 48:516–535.